

On computing the analytic-signal backbone of the unforced harmonic oscillator

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ABSTRACT

Backbone curves are usually obtained by analyzing nonlinear oscillators (e.g., the Duffing equation) whereas the backbones of linear oscillators are incidental by-products of such studies. Nevertheless, we focus on the harmonic oscillator, a linear, homogeneous constant-coefficient second-order ordinary differential equation. The backbone of the sinusoidally forced harmonic oscillator is well known, but the backbone of the unforced harmonic oscillator seems to have received essentially no attention. Since the 1980s, various computerized methods of backbone extraction have been under development, one of which is based on analytic-signal theory, a subfield of digital signal processing that involves Hilbert transform computations. By focusing on linear-theory backbones, a prerequisite to nonlinear analysis, we aim to pinpoint some of the theoretical and computational issues that arise when dealing with analytic-signal backbones. This paper examines the Mirror Periodic Method (MPM) as a practical way to reduce end effects in Hilbert transform computations and approximate analytic-signal backbones of unforced harmonic oscillators with light damping (less than 10 percent of critical). Using numerical examples, we demonstrate that MPM works well with both displacement and acceleration data, even for short signals, meaning just one or two damped cycles of oscillation. We then apply MPM techniques to accelerometer data that were recorded as part of earthquake-engineering teaching demonstrations involving a small-scale laboratory model of a linear single-degree-of-freedom system.

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1. Introduction

Computerized methods of nonlinear dynamical systems identification often involve the extraction of backbone curves from digitized signals. In addition to the Hilbert transform approach to backbone extraction [1,2], there are other techniques which are not discussed here since they are adequately covered elsewhere; see for example [3–6] and the references therein. Underpinning nonlinear analysis is linear theory, exemplified by the harmonic oscillator, which is the focus of this paper.

An example of a nonlinear oscillator whose response characteristics (i.e., solutions) are well known is the sinusoidally forced Duffing oscillator [7,8], which can be written in the form

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x + \alpha x^3 = B\cos(\beta t), \text{ subject to } x(0) = x_0, \dot{x}(0) = \dot{x}_0 \quad (1)$$

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Nomenclature

$a(t)$	Instantaneous amplitude of displacement signal $x(t)$
$A(t)$	Instantaneous amplitude of acceleration signal $\ddot{x}(t)$
B	Amplitude of sinusoidal excitation in Duffing equation
f_n	Natural frequency (Hz)
f_s	Sampling rate (Hz) of accelerometer data
$j = \sqrt{-1}$	Imaginary unit
K	Integer number of points per damped cycle
N	Number of points in test signal
$u(t)$	The unit step function
$x(t)$	Real signal, displacement
$\tilde{x}(t)$	Hilbert transform of $x(t)$
$z(t) = x(t) + j\tilde{x}(t)$	Analytic signal associated with $x(t)$
α	Cubic stiffness parameter in Duffing equation
β	Frequency (rad/s) of sinusoidal excitation in Duffing equation
$\omega(t)$	Instantaneous frequency of $x(t)$
$\omega_n = 2\pi f_n$	Undamped linear oscillator frequency (rad/s)
$\Omega(t)$	Instantaneous frequency of $\ddot{x}(t)$
$\phi(t)$	Instantaneous phase of $x(t)$
ζ	Damping ratio ($\zeta = 1$ is critically damped)

where $x = x(t)$ denotes the response (displacement) as a function of time t , ω_n is the undamped linear oscillator frequency (rad/s), ζ is the damping ratio ($\zeta = 1$ is critically damped), α is the cubic stiffness parameter, B is the excitation (or forcing) amplitude and β is the sinusoidal excitation frequency (rad/s).

If $\alpha = 0$ and $\zeta > 0$, Eq. (1) reduces to a sinusoidally forced harmonic oscillator (i.e., a linear nonhomogeneous second-order ordinary differential equation with constant coefficients) whose solutions can be described in physical terms as the linear superposition of the transient response (determined solely by $x(0)$ and $\dot{x}(0)$) and the steady-state response (determined solely by the sinusoidal excitation). As t goes to $+\infty$, the transient response becomes negligible compared to the steady-state response. Fig. 1 is a plot of the well-known amplification (or magnification) factor,

$$M\left(\frac{\beta}{\omega_n}\right) = \frac{1}{\sqrt{\left(1 - \left(\frac{\beta}{\omega_n}\right)^2\right)^2 + 4\zeta^2 \left(\frac{\beta}{\omega_n}\right)^2}} \quad (2)$$

which equals the amplitude of the steady-state response at forcing frequency β divided by the static value at $\beta = 0$ (e.g., see Eq. (5.2.9) in [7] or Eq. (1.12) in [8]).

The vertical straight line at $\beta = \omega_n$ in Fig. 1 is the linear-theory backbone, which corresponds to the infinite value of $M(1)$ in the limit as ζ goes to zero. Fig. 2(a) shows three steady-state backbones for the Duffing equation corresponding to: (1) $\alpha = 0$, which is the linear-theory backbone shown in Fig. 1; (2) $\alpha > 0$, called a “hardening” response backbone; (3) $\alpha < 0$, called a “softening” response backbone. Generally speaking, extracting a backbone from a digitized response signal and then classifying it according to shape is a key step in the process of system identification. For example, assuming steady-state Duffing oscillator response, extraction of a straight vertical backbone would indicate linear response, while a backbone bending to the right would indicate hardening response, and a backbone bending to left would indicate softening response. Fig. 2(b) shows idealized analytic-signal backbones corresponding to those in Fig. 2(a), the only difference being the word “instantaneous” in the labels of the ordinates and the abscissae. Feldman [9] illustrates backbones associated with various types of nonlinearity including (but not necessarily limited to) hardening, softening, backlash, pre-compressed springs, bilinear, and impact. In principle, analytic-signal backbone analysis has the potential for being a classifier of a nonlinear dynamical system using output signals only. By focusing on linear-theory backbones, a prerequisite to nonlinear problems, this paper aims to pinpoint some of the theoretical and computational issues that arise when dealing with analytic-signal backbones.

2. Structure of this paper

Rather than tackling nonlinear oscillators, this paper focuses on linear oscillators, specifically the unforced harmonic oscillator with damping,

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = 0, \quad \text{subject to } x(0) = x_0, \quad \dot{x}(0) = \dot{x}_0. \quad (3)$$

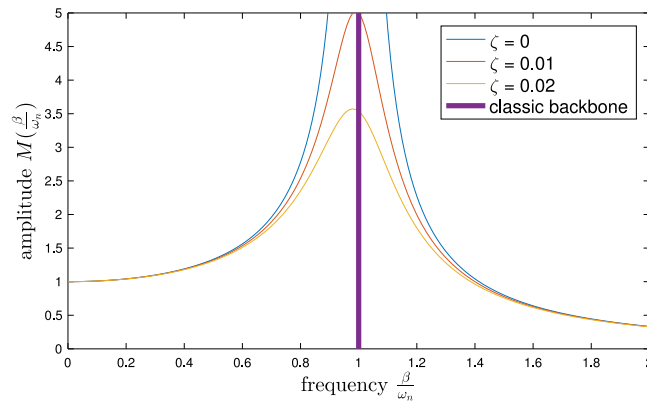


Fig. 1. Linear steady-state response for $\zeta = 0, 0.1, 0.2$ and the linear-theory backbone.

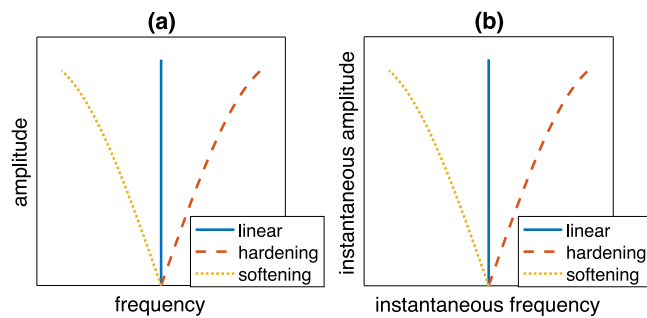


Fig. 2. (a) Steady-state backbones of the Duffing oscillator; (b) Idealized analytic-signal backbones: (1) $\alpha = 0$, linear theory (see Fig. 1); (2) hardening, $\alpha > 0$; (3) softening, $\alpha < 0$.

Table 1

Maximum values of damping ratio for different structural types; see Table A17.5 in [12].

Type	Steel (Welded)	Steel (Bolted)	Concrete (Reinforced)	Concrete (Prestressed)	Masonry (Reinforced)	Wood Trusses	Wood Frames
ζ (%)	2.0	4.0	5.0	3.0	6.0	9.0	7.0

The aim of this paper is to summarize issues that are commonly presented in a nonlinear context, which tends to obscure subtle details that become clearer in a purely linear context. The preceding Section 1 outlined classical backbone theory and pointed out conceptual differences between the linear steady-state backbone and the linear unforced (free vibration) backbone, especially when the damping parameter ζ is non-zero. Section 3 defines the analytic-signal backbone in terms of the Hilbert transform (HT), and discusses various mathematical and theoretical difficulties that arise when the HT is applied to damped solutions of Eq. (3). For example, the fact that the Hilbert transform of an exponentially decaying sinusoid does not exist is discussed. Section 4 then points out that because of such difficulties, the quadrature-signal backbone is, by convention, the preferred alternative to the analytic-signal backbone; it then discusses the Mirror Periodic Method (MPM) [10], with and without filtering, as a practical approach for computing the quadrature-signal backbone of the unforced harmonic oscillator with light damping, defined here as $0 \leq \zeta \leq 0.1$ (critical damping is $\zeta = 1$). Earthquake engineering references such as [11,12] indicate that this is an important range for many structures. Maximum values of ζ in the light damping range are summarized in Table 1 and numerical results presented in this paper are restricted to this range.

In Sections 5 and 6, we show that MPM works well with both displacement and acceleration signals, even for short signals, meaning just one or two damped cycles of oscillation. In Section 7, we apply MPM techniques to accelerometer data that were recorded as part of an earthquake-engineering teaching demonstration involving a small-scale laboratory model of a linear single-degree-of-freedom (SDOF) system. Although multi-degree-of-freedom problems are of interest, we do not discuss this subject since many techniques for decomposing a signal into “modes” are discussed in the literature under various names such as “Empirical Mode Decomposition (EMD)”, “Hilbert Vibration Decomposition (HVD)”, and “Analytical Mode Decomposition (AMD)”.

3. Background on the analytic-signal backbone of the unforced harmonic oscillator

In the framework of analytic-signal theory, a subfield of Digital Signal Processing (DSP), let $x(t)$ be a real signal and $\tilde{x}(t)$ denote its Hilbert transform

$$\tilde{x}(t) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau \quad (4)$$

where P indicates the Cauchy principal value. The analytic signal associated with $x(t)$ is the complex signal $z(t) = x(t) + j\tilde{x}(t)$. In polar notation,

$$z(t) = a(t)e^{j\phi(t)} \quad (5)$$

where $a(t) = \sqrt{x^2(t) + \tilde{x}^2(t)} \geq 0$ is the instantaneous amplitude, $\phi(t) = \tan^{-1} \left(\frac{\tilde{x}(t)}{x(t)} \right)$ is the instantaneous phase, and $\omega(t) = \frac{d\phi}{dt}$ is the instantaneous frequency. The phrase “analytic-signal backbone” (ASB) refers to the parametric plot $a(t)$ versus $\omega(t)$.

As a test signal and without loss of generality, consider

$$x(t) = \frac{e^{-\zeta t}}{\sqrt{1 - \zeta^2}} \cos \left[\sqrt{1 - \zeta^2} t + \tan^{-1} \frac{-\zeta}{\sqrt{1 - \zeta^2}} \right] \quad (6)$$

which is the solution of the unforced harmonic oscillator problem, Eq. (3), with $x(0) = 1$, $\dot{x}(0) = 0$ and $\omega_n = 1$.

At the outset, it should be noted that the Hilbert transform (HT) of an exponentially decaying sinusoid, such as the test signal, does not exist because, if $\zeta > 0$, the integrand becomes infinite as t goes to $-\infty$. Alternatively, if the test signal $x(t)$ is assumed to be zero for $t < 0$ (i.e., an initial value problem), then the HT of $u(t)x(t)$, where $u(t)$ denotes the unit step function, does not exist at $t = 0$, because the HT of $u(t)x(t)$ has a logarithmic singularity there, caused by the jump discontinuity of $u(t)$. Loughlin [13] thoroughly discussed $u(t)\cos(t)$, which is the solution of the undamped initial-value problem. But even with positive damping $\zeta > 0$, the jump discontinuity of $u(t)x(t)$ at $t = 0$ still causes its HT to have a logarithmic singularity at $t = 0$.

Before seeking the analytic-signal backbone (ASB) of the test signal, we note that $x(t)$ in Eq. (6) has the mathematical form $a(t)\cos(\phi(t))$, so it is tempting to categorize $x(t)$ as an amplitude-modulated (AM) signal with constant instantaneous frequency $\omega(t) = \sqrt{1 - \zeta^2}$ and time-varying instantaneous amplitude $a(t) = \frac{e^{-\zeta t}}{\sqrt{1 - \zeta^2}}$. In other words, the ASB of $x(t)$ might be assumed to be a vertical straight-line AM backbone. However, identifying a real signal's instantaneous amplitude and instantaneous frequency modulation characteristics is an ill-posed problem, because the number of amplitude-phase pairs that can generate a given real signal $x(t)$ is infinite, as has been pointed out many times, [13–17].

For example, as Picinbono makes clear in the Introduction of Picinbono [17], $x(t) = \cos(\theta(t))$ is an equally valid way to express the test signal $x(t)$ because $|x(t)| \leq 1$ for all $t \geq 0$. If written this way, it is tempting to categorize $x(t)$ as a frequency-modulated (FM) signal with constant instantaneous amplitude $a(t) = 1$ and time-varying instantaneous frequency $\omega(t) = \frac{d\theta}{dt}$. In other words, the ASB of $x(t)$ might be assumed to be a horizontal straight-line FM backbone, not a vertical straight-line AM backbone, neither of which can be stated categorically without invoking additional conditions based on (for example) physical reasoning.

In principle, analytic-signal theory offers a means to circumvent the ill-posedness (nonuniqueness) issue. After all, the Hilbert transform (HT) of the test signal exists if $\zeta = 0$ since the HT of $\cos(t)$ is $\sin(t)$. Hence, the ASB of the undamped $x(t)$ is simply the (1, 1) point in the amplitude–frequency plane. However, the (1, 1) point is neither AM nor FM, nor is it the linear-theory backbone. The linear-theory backbone in Fig. 1 is an abstraction derived from a different physical model than Eq. (3). It is obtained by analyzing the forced harmonic oscillator with sinusoidal excitation. Its family of amplitude–frequency response curves depends on ζ , and the linear-theory backbone corresponds to the steady-state case with $\zeta = 0$. Due to resonance (i.e., secular terms such as $t \sin(t)$; see Eq. (5.2.11) in [7]), the linear-theory backbone is a semi-infinite straight-line vertical (AM) backbone that passes through the (1,1) point in the steady-state amplitude–frequency plane, increasing its amplitude as time increases. In other words, it is a “resonance backbone”.

Instead of a forced vibration problem, the test signal corresponds to an unforced (free vibration) problem, which explains why the ASB of the undamped $x(t)$, being the (1, 1) point in the amplitude–frequency plane, is not a linear-theory backbone. In other words, the test signal corresponds to transient response whereas the linear-theory backbone corresponds to steady-state response. Indeed, it is not obvious that the amplitude–frequency plane of the steady-state backbones illustrated in Fig. 2(a), should be physically equivalent to the instantaneous amplitude–frequency plane defined by analytic-signal theory and illustrated in Fig. 2(b). Nevertheless, the semi-infinite straight-line AM backbone in Fig. 1 is presumed to be the “benchmark” for all linear-theory backbone analysis, whether forced (steady-state) or unforced (transient) vibrations are being analyzed; see [8], page 147.

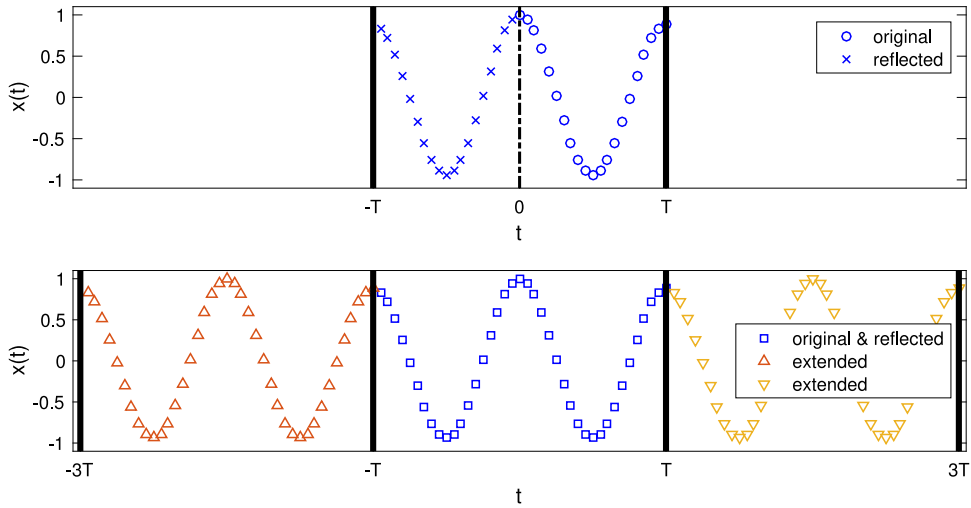


Fig. 3. Illustration of how an original signal, having zero slopes at both end points, is reflected to form its even periodic extension. Dashed and solid vertical lines indicate “mirrors”.

4. The quadrature signal and approximate analytic-signal backbones

Theoretical difficulties such as those summarized in the previous section have provoked various DSP approaches that (eventually) produced approximate analytic-signal backbones for unforced harmonic oscillator problems. First of all, the signal that is 90 degrees out of phase with the test signal, called the quadrature signal,

$$x_q(t) = \frac{e^{-\zeta t}}{\sqrt{1-\zeta^2}} \sin \left[\sqrt{1-\zeta^2} t + \tan^{-1} \frac{-\zeta}{\sqrt{1-\zeta^2}} \right] \quad (7)$$

is generally taken to be the best approximation to the Hilbert transform [18]. In other words, the quadrature-signal backbone, with instantaneous amplitude $a(t) = \frac{e^{-\zeta t}}{\sqrt{1-\zeta^2}}$ and instantaneous frequency $\omega(t) = \sqrt{1-\zeta^2}$ is conventionally deemed to be the analytic-signal backbone of the unforced harmonic oscillator (without loss of generality $\omega_n = 1$).

Many numerical methods for computing the Hilbert transform are discussed in the literature (e.g., [19–23]). Regardless of method, difficulties arise because the domain of integration in Eq. (4) extends from $-\infty$ to $+\infty$ whereas only finite-length discrete-time signals are normally available, so approximations at both ends must be considered. As we will show, the Mirror Periodic Method (MPM) [10] works well for unforced harmonic oscillators with light damping.

Consistent with MPM applied to Eq. (6), “mirrors” are placed as follows: a mirror at $t = 0$, creating a symmetric (even) signal with (nearly) twice as many sample points as the original signal, and then mirrors at both ends of the symmetrized signal, thus creating an MPM signal that is an even periodic extension with zero slopes, $\dot{x}(t) = 0$, at both ends of the original signal. As expected, examination of the velocity,

$$\dot{x}(t) = -\frac{e^{-\zeta t}}{\sqrt{1-\zeta^2}} \sin \left(\sqrt{1-\zeta^2} t \right), \quad (8)$$

shows that the zero slope condition is satisfied at an infinity of times equal to integer multiples of $\frac{\pi}{\sqrt{1-\zeta^2}}$.

In other words, since the HT of a finite-length signal is problematic to compute, $\tilde{x}(t)$ will be approximated by the HT of the even periodic extension with zero slopes at both ends of the original signal. This scheme has the benefit of eliminating logarithmic singularities at the two end times when compared to computing the HT with zero padding outside the signal, but at the cost of nearly doubling the number of samples. In practice, this cost can be eliminated by accounting for even periodicity of the mirrored signal (see Eq. (6.86) in [24], subsection 6.7). In the limit as ζ goes to zero, the scheme approaches the exact result for the test signal, namely the (1, 1) point in the instantaneous amplitude–frequency plane, so it is reasonable to expect it will give acceptable (not perfect) results for small ζ . Our studies show this happens for unforced harmonic oscillators with light damping, $0 \leq \zeta \leq 0.1$.

Given a signal of length N , some care is needed to construct a proper MPM signal. For the test signal Eq. (6), one way to do this is by reversing the sample points of the original signal while omitting the first and last points (presuming these two points are at times when $\dot{x}(t) = 0$), thus creating a signal of length $2N - 2$ as in Fig. 3 (see Figure 1 in [10]).

Figs. 4–6 present computational results for two damping ratios, $\zeta = 0.02$ and $\zeta = 0.1$. For fixed ζ , the series of approximate analytic-signal backbones exhibit a pattern that can be described as oscillating AM backbones because, as the

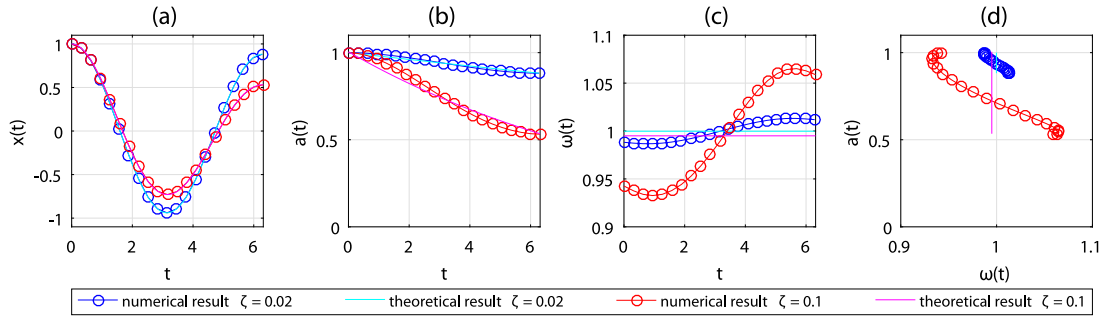


Fig. 4. One damped cycle: (a) Test signals $x(t)$ for $\zeta = 0.02$ and $\zeta = 0.1$ with $K = 21$ points per damped cycle; (b) instantaneous amplitude $a(t)$; (c) instantaneous frequency $\omega(t)$; (d) backbone $a(t)$ vs. $\omega(t)$.

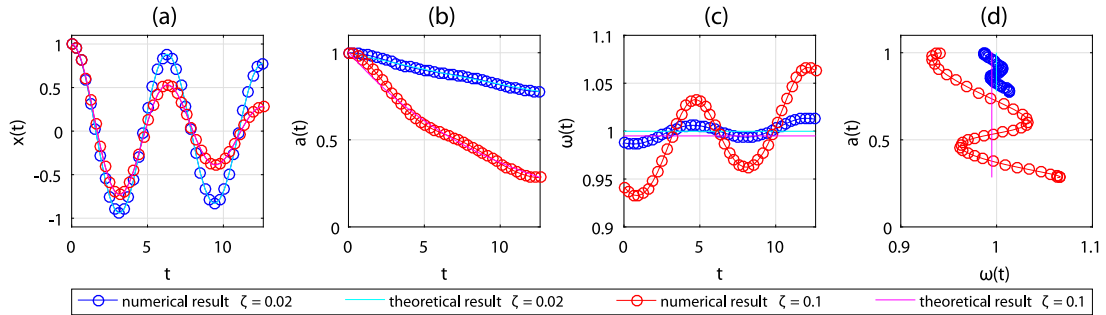


Fig. 5. Two damped cycles: (a) Test signals $x(t)$ for $\zeta = 0.02$ and $\zeta = 0.1$ with $K = 21$ points per damped cycle; (b) instantaneous amplitude $a(t)$; (c) instantaneous frequency $\omega(t)$; (d) backbone $a(t)$ vs. $\omega(t)$.

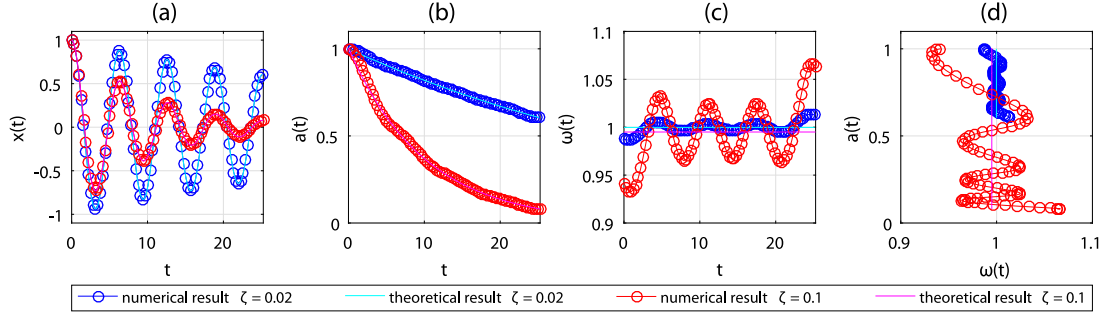


Fig. 6. Four damped cycles: (a) Test signals $x(t)$ for $\zeta = 0.02$ and $\zeta = 0.1$ with $K = 21$ points per damped cycle; (b) instantaneous amplitude $a(t)$; (c) instantaneous frequency $\omega(t)$; (d) backbone $a(t)$ vs. $\omega(t)$.

number of damped cycles increases (i.e., as time increases), the backbones decrease in the AM direction, while oscillating in the FM direction with a relatively small, apparently bounded magnitude. This happens because the test signal $x(t)$ decays exponentially in time with a decay rate that is first order in ζ , while $x(t)$ simultaneously oscillates in time with a frequency shift that is relatively small, being only second order in ζ .

Figs. 4–6 were created using proper MPM signals and MATLAB's "hilbert.m" which is a HT computational method based on the fast Fourier transform (FFT) [25]. Proper MPM signals are well-suited to FFT computations. To reduce discretization errors (aliasing, leakage, etc.), the test signals were carefully sampled at an integer number K points per damped cycle, where one period of a damped cycle is $\frac{2\pi}{\sqrt{1-\zeta^2}}$, and the first and last points are at times when the velocity $\dot{x}(t) = 0$. The three figures correspond to 1, 2 and 4 damped cycles, respectively. Enforcing an integer number K sample points per damped cycle is equivalent to placing mirrors at times when $\dot{x}(t) = 0$.

For comparison purposes, all (b) subplots include exponentially decaying curves corresponding to $\frac{e^{-\zeta t}}{\sqrt{1-\zeta^2}}$, and all (c) subplots include straight lines corresponding to $\sqrt{1-\zeta^2}$ which is the damped frequency. Accordingly, the backbones in the (d) subplots include two vertical straight lines corresponding to $\zeta = 0.02$ and $\zeta = 0.1$. Although these two

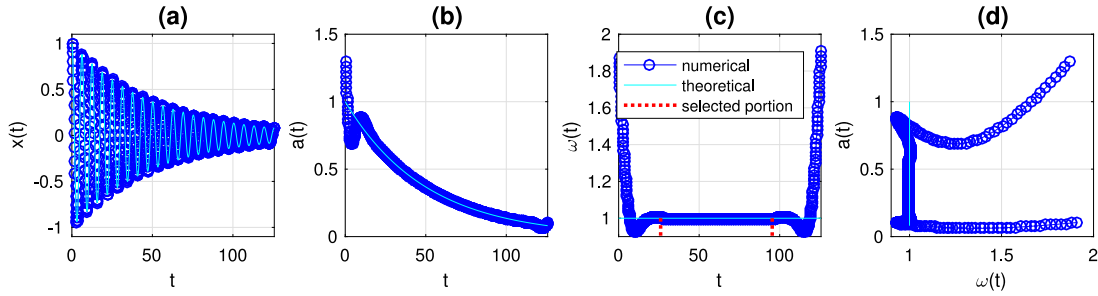


Fig. 7. Results for 20 damped cycles, obtained using HVD with $n = 1$, $f_p = 0.02$: (a) Test signal $x(t)$ with $N = 801$ points, $\zeta = 0.02$; (b) instantaneous amplitude $a(t)$; (c) instantaneous frequency $\omega(t)$; (d) backbone $a(t)$ vs. $\omega(t)$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

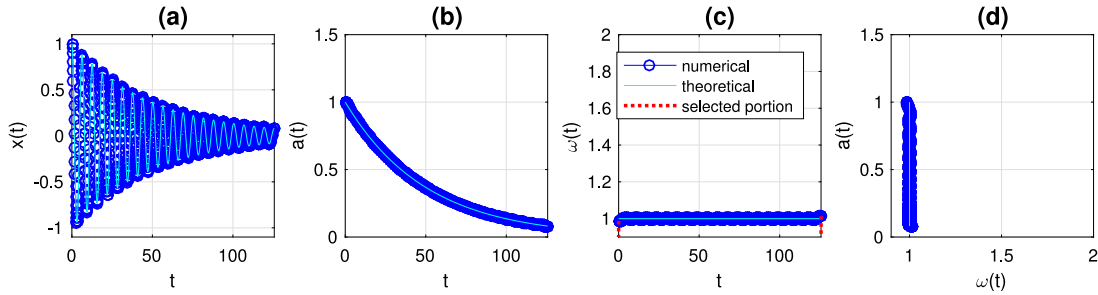


Fig. 8. Results for 20 damped cycles, obtained using 1600-point MPM signal and “hilbert.m” (MATLAB): (a) Original test signal $x(t)$ with $N = 801$ points, $\zeta = 0.02$; (b) instantaneous amplitude $a(t)$; (c) instantaneous frequency $\omega(t)$; (d) backbone $a(t)$ vs. $\omega(t)$.

straight-line AM backbones resemble the linear-theory backbone, they are actually quite different, being instead the quadrature-signal backbones of the test signal.

It should be noted that separate Hilbert transform computations with the method in [23] were conducted with the same proper MPM signals, producing results (not reported here) similar to Figs. 4–6, although that method is not based on the Fourier transform identity implemented in MATLAB’s “hilbert.m”. In other words, the oscillatory nature of the backbones in the (d) subplots of Figs. 4–6 persisted, regardless of HT computational method.

5. Filtering and end effects

Unless filtering is introduced, Hilbert transform computations with proper MPM signals produce analytic-signal backbones that oscillate, which is undesirable when compared to the quadrature-signal backbones in the (d) subplots of Figs. 4–6. It is worth noting that a 1998 survey [6] states that the Hilbert transform approach to backbone extraction requires significant filtering efforts. Also, Vakman [15] discusses “oscillating APF” (APF denotes Amplitude, Phase and Frequency) and filtering when doing analytic-signal analysis.

One of many possible filtering approaches is the Hilbert “transformer” (e.g., [26]) based on the Remez exchange algorithm, mentioned in [1] with online software available as part of Hilbert Vibration Decomposition (HVD). We used relevant parts of this software (dated 03-Dec-2010) without modification. A limitation (stated in the HVD software comments) is that the signal length must be 691 points or more (in contrast to the results in the preceding section). Consequently for the next set of results, the number of damped cycles was increased to 20 and the sampling frequency was increased to 40 (i.e., equivalent to $K = 41$ points per damped cycle), producing a test signal with $N = 801$ points. The damping ratio was set to $\zeta = 0.02$. HVD results for this 801-point signal are presented in Fig. 7 which shows end effects, meaning errors near the two ends. Next, a proper 1600-point MPM signal was created from the 801-point test signal. Fig. 8 shows unfiltered MPM results, obtained by using the same scheme as in Figs. 4–6. As expected, the MPM backbone oscillates but end effects are small relative to the HVD results.

In an attempt to reduce the HVD end effects in Fig. 7, the 1600-point MPM signal that produced Fig. 8 was subjected to HVD, resulting in considerable improvement at one end corresponding to $t = 0$, as Fig. 9 shows. Finally, the 1600-point MPM signal was converted into a proper 3198-point MPM signal, called MPM2 and shown in Fig. 10, which was subjected to HVD, resulting in Fig. 11 which shows considerable improvement at both ends.

The captions of Figs. 7, 9 and 11 state that those HVD results were obtained with $n = 1$ (one mode), and filtering parameter $f_p = 0.02$ which refers to two user-defined values of f_p in two HVD MATLAB scripts, “synchdem.m” and “lpf.m”. However, it should be noted that a third HVD MATLAB script, “hilbfir.m”, contains a built-in filtering parameter $f_p = 0.01$

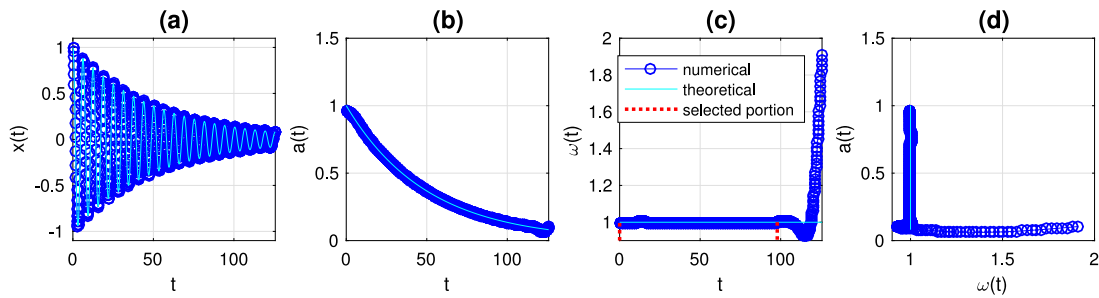


Fig. 9. Results for 20 damped cycles, obtained using 1600-point MPM signal and HVD with $n = 1$, $f_p = 0.02$: (a) Original test signal $x(t)$ with $N = 801$ points, $\zeta = 0.02$; (b) instantaneous amplitude $a(t)$; (c) instantaneous frequency $\omega(t)$; (d) backbone $a(t)$ vs. $\omega(t)$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

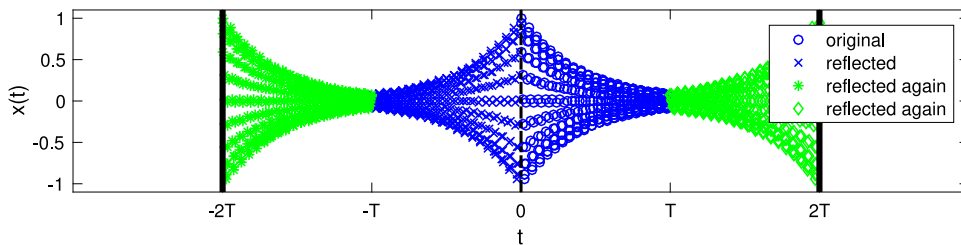


Fig. 10. Illustration of how original test signal $x(t)$ with $N = 801$ points was converted into a 1600-point MPM signal, which in turn was converted into a 3198-point signal, MPM2.

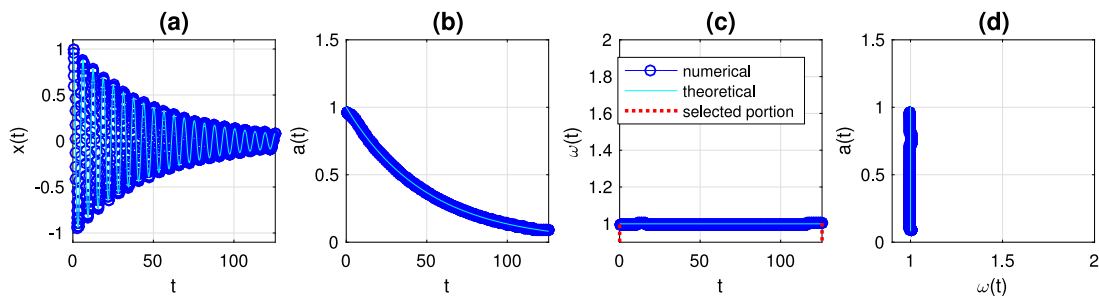


Fig. 11. Results for 20 damped cycles, obtained using 3198-point MPM2 signal and HVD with $n = 1$, $f_p = 0.02$: (a) Original test signal $x(t)$ with $N = 801$ points, $\zeta = 0.02$; (b) instantaneous amplitude $a(t)$; (c) instantaneous frequency $\omega(t)$; (d) backbone $a(t)$ vs. $\omega(t)$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

RMS errors of the instantaneous frequency compared to the damped frequency $\sqrt{1 - \zeta^2}$. † Note: $f_p = 0.01$ is unchanged in “hilbfir.m”; f_p in “synchdem.m” and “lpf.m” were set as specified below.

In “synchdem.m”, “lpf.m”	HVD	MPM	MPM + HVD	MPM2 + HVD
$f_p = 0.02^\dagger$ entire signal	0.1557	0.0026	0.1122	0.001
$f_p = 0.02^\dagger$ selected portion	0.001	0.0026	0.0008	0.001
$f_p = 0.01$ entire signal	0.2267	0.0026	0.1637	0.0199
$f_p = 0.01$ selected portion	0.0199	0.0026	0.0197	0.0199

that was unchanged. HVD computations were also conducted with $f_p = 0.01$ in all three HVD MATLAB scripts but these results are inferior, as can be seen in Table 2 which gives the root-mean-square (RMS) error of the instantaneous frequency compared to the theoretical damped frequency $\sqrt{1 - \zeta^2}$. RMS values were computed for the entire signal, and for the “selected portion” of the signal between the red lines in the (c) subplots. The results in Table 2 indicate that end effects can be reduced considerably by combining MPM signals and hilbert.m, or by combining MPM2 signals with a good filtering scheme such as HVD.

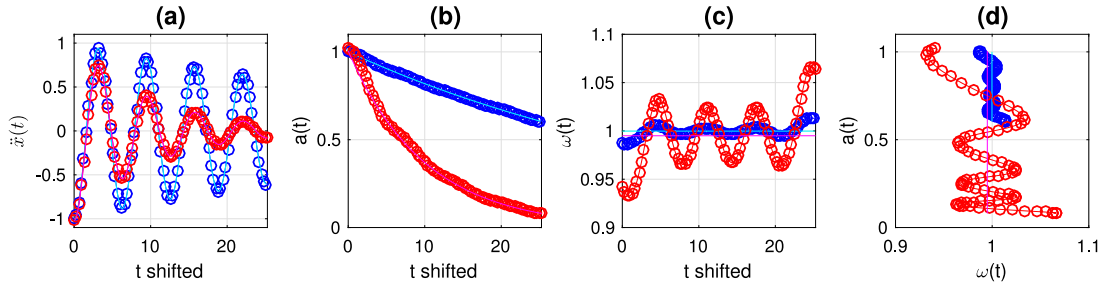


Fig. 12. Four cycles of acceleration with $\zeta = 2, 10\%$.

6. Acceleration signals and the mirror periodic method

Accelerometers are normally used to measure structural vibrations, which leads to the practical question of how well do mirror-periodic methods (MPM) work with acceleration signals instead of displacement signals. In cases of light damping, we find that extracting backbones while using MPM and MPM2 signals works just as well with an acceleration signal as it does with the corresponding displacement (test) signal having the same value of ζ . One way to see this is by taking the second derivative of $x(t)$ in Eq. (6), yielding the corresponding acceleration signal,

$$\ddot{x}(t) = -\frac{e^{-\zeta t}}{\sqrt{1-\zeta^2}} \cos \left[\sqrt{1-\zeta^2} t + \tan^{-1} \frac{\zeta}{\sqrt{1-\zeta^2}} \right]. \quad (9)$$

Comparison of Eqs. (6) and (9) shows that they differ by just two sign changes: one is the minus sign after the equal sign, and one is in the phase (the argument of arctan). By itself, the first sign change would not change the backbone at all, while the sign change in the phase should cause small changes so long as ζ is small. Numerical studies confirm this, showing that, if $0 \leq \zeta \leq 0.1$, the backbone extracted from an acceleration signal is nearly the same as the backbone extracted from the displacement signal with the same value of ζ . Fig. 12 shows results for four damped cycles of the acceleration signal with $\zeta = 0.02, 0.1$.

Comparing Fig. 12 and the corresponding results in Fig. 6, there is an obvious difference in subplot (a) where the sign of the acceleration signal is opposite to the sign of the corresponding displacement signal. Also, the starting times in subplots (a), (b) and (c) actually should be non-zero because the zero slopes of the acceleration signal Eq. (9) occur at an infinite set of times,

$$t = \frac{k\pi - \tan^{-1} \left(\frac{2\zeta\sqrt{1-\zeta^2}}{1-2\zeta^2} \right)}{\sqrt{1-\zeta^2}}, \quad k = 0, \pm 1, \pm 2, \dots, \quad (10)$$

none of which are zero when $\zeta > 0$. Instead, the abscissa are labeled “t shifted” because the starting times are set equal to zero while maintaining the same sampling rate, which is appropriate because Eq. (3) is a linear time-invariant (LTI) system, so its displacement signals as well as its acceleration signals can be compared point by point by shifting time and accounting for the undamped frequency which is $\omega_n = 1$ in Eqs. (6) and (9). After accounting for LTI and the undamped frequency, the instantaneous amplitudes extracted from displacement and acceleration signals can be compared, resulting in root-mean-square errors of 6.4×10^{-4} and 9.3×10^{-3} for $\zeta = 2\%, 10\%$, respectively. Even more interesting are the root-mean-square errors of the instantaneous frequencies which are zero numerically (within round-off error which is about 14 figures) for both values of ζ .

One way to estimate the damping ratio ζ is to equate the damping coefficient $\zeta\omega_n$ in Eq. (3) to the instantaneous damping coefficient $h_0(t)$ in Eq. (6) of Feldman [9], yielding the following expression for the instantaneous damping ratio

$$\zeta(t) = \underbrace{-\frac{1}{\omega_n} \frac{\dot{A}(t)}{A(t)}}_{\zeta_1(t)} - \underbrace{\frac{1}{\omega_n} \frac{\dot{\Omega}(t)}{2\Omega(t)}}_{\zeta_2(t)} \quad (11)$$

where $A(t)$ and $\Omega(t)$ denote the instantaneous amplitude and frequency, respectively, of an acceleration signal. In theory, the second term $\zeta_2(t)$ should be zero for a linear oscillator, but numerically we find it contributes significantly to the noisiness of the instantaneous damping ratio, especially near the two ends of the signal. For this reason, we neglect the second term and consider only $\zeta_1(t)$ when estimating the damping ratio, as illustrated next.

Fig. 13 shows MPM results for 20 damped cycles of the acceleration signal Eq. (9) with $\zeta = 2\%$ and 40 points per damped cycle (the same sampling rate as Fig. 8) while Figs. 14 and 15 show corresponding HVD and MPM2 + HVD results, respectively.

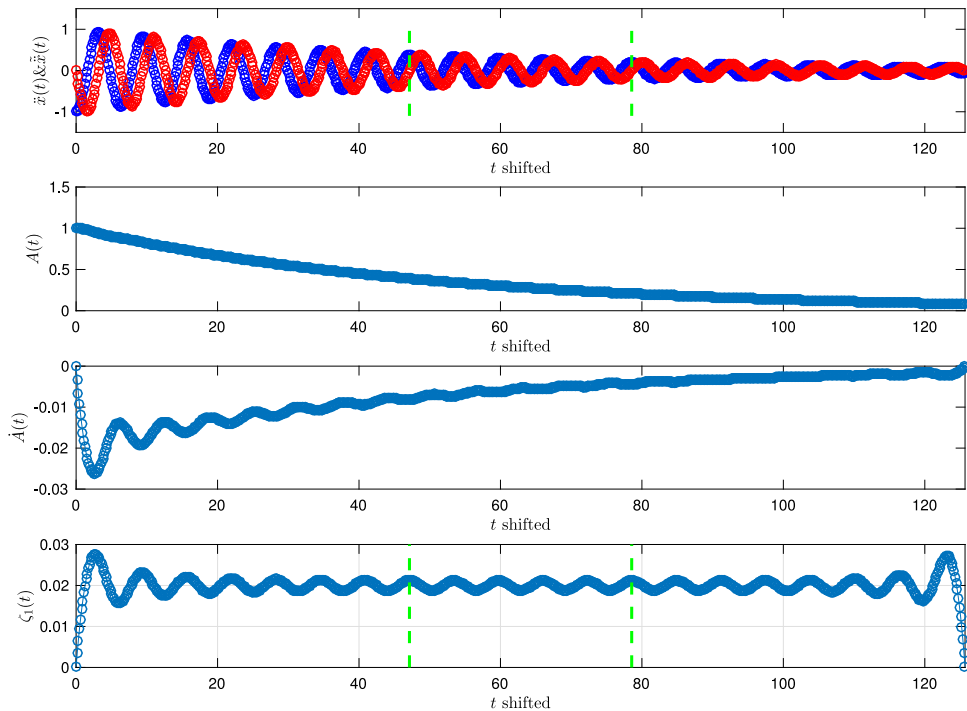


Fig. 13. Example of estimating ζ by averaging $\zeta_1(t)$ over the middle 5 damped cycles of acceleration signal (out of 20 cycles); using MPM+hilbert.m; Table 3.

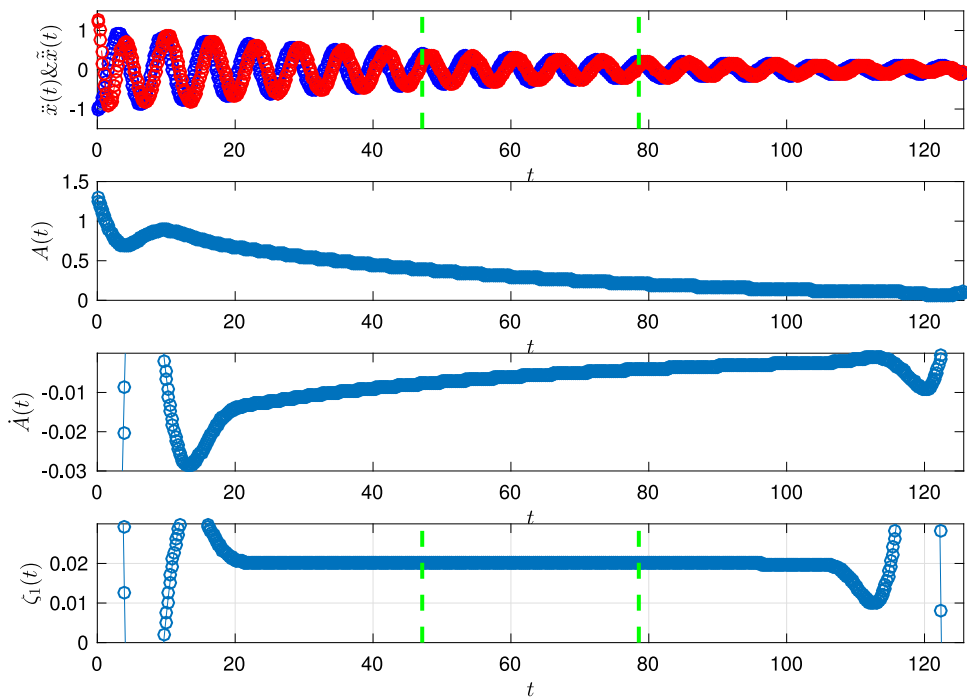


Fig. 14. Example of estimating ζ by averaging $\zeta_1(t)$ over the middle 5 damped cycles of acceleration signal (out of 20 cycles); using HVD with $n = 1$, $f_p = 0.02$ (user-defined) in “synchdem.m” and “lpf.m”, $f_p = 0.01$ (built-in) in “hilbfir.m”, Table 3.

As the $\zeta_1(t)$ signal in Fig. 13 illustrates, unfiltered MPM results are oscillatory (as expected). On the other hand, the HVD results in Fig. 14 are filtered and thus are smooth, but only in the central portion of the HVD $\zeta_1(t)$ signal. Table 3

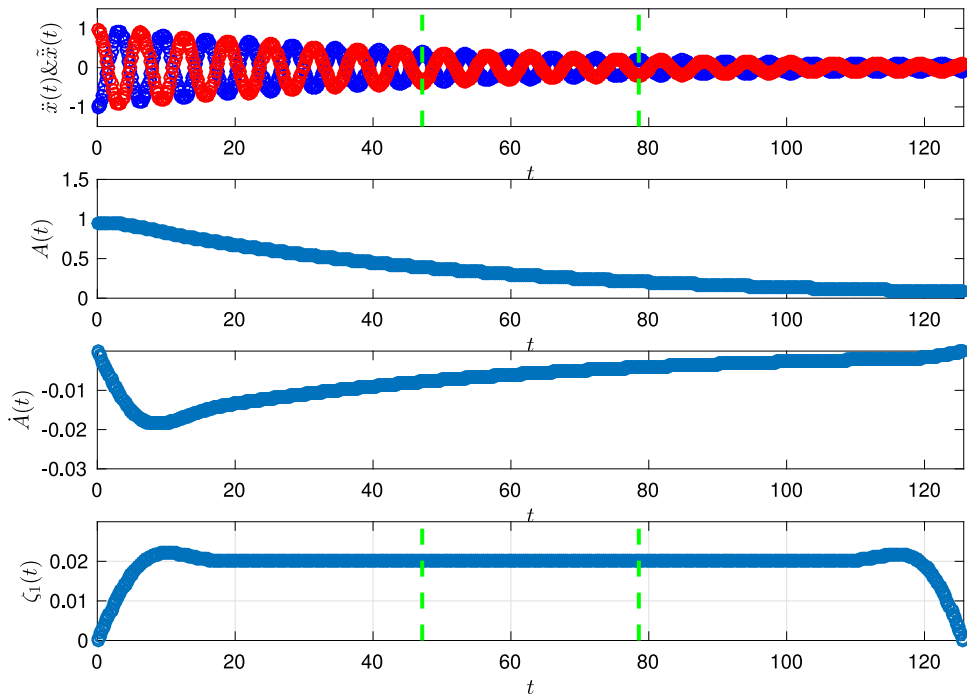


Fig. 15. Example of estimating ζ by averaging $\zeta_1(t)$ over the middle 5 damped cycles of acceleration signal (out of 20 cycles); using MPM2 + HVD with $n = 1$, $f_p = 0.02$ (user-defined) in “synchdem.m” and “lpf.m”, $f_p = 0.01$ (built-in) in “hilbfir.m”, Table 3.

Table 3

Unsigned relative error of damping ratio estimates, obtained by averaging $\zeta_1(t)$ over the middle damped cycles (out of 20 cycles); the exact ζ value is 2%; $n = 1$, $f_p = 0.02$ (user-defined) in “synchdem.m” and “lpf.m”, $f_p = 0.01$ (built-in) in “hilbfir.m”.

No. of cycle	MPM+hilbert.m	HVD	MPM+HVD	MPM2+HVD
1	-1.470×10^{-1}	-1.646×10^{-4}	-1.646×10^{-4}	-1.646×10^{-4}
3	-5.040×10^{-2}	-1.644×10^{-4}	-1.646×10^{-4}	-1.646×10^{-4}
5	-3.097×10^{-2}	-1.630×10^{-4}	-1.660×10^{-4}	-1.646×10^{-4}
7	-2.288×10^{-2}	$+1.411 \times 10^{-4}$	-3.171×10^{-4}	-1.678×10^{-4}
9	-1.872×10^{-2}	$+4.234 \times 10^{-3}$	-2.791×10^{-3}	-1.435×10^{-4}
11	-1.655×10^{-2}	-7.924×10^{-3}	$+1.638 \times 10^{-2}$	$+2.154 \times 10^{-4}$
13	-1.584×10^{-2}	-4.622×10^{-2}	$+1.090 \times 10^{-1}$	$+1.439 \times 10^{-2}$
15	-3.113×10^{-2}	-6.887×10^{-1}	$+6.195 \times 10^{-1}$	-3.363×10^{-2}
17	-6.342×10^{-2}	-9.768×10^{-1}	$+1.293 \times 10^0$	-8.147×10^{-1}

gives estimates of the damping ratio ζ , obtained by averaging the $\zeta_1(t)$ signals over a number of cycles, starting in the center of the signals.

7. Analysis of accelerometer data from small-scale laboratory test

Accelerometer data were recorded as part of an earthquake-engineering teaching demonstration involving a small-scale laboratory test of a linear SDOF system, corresponding to a single-story shear-beam model. Photos showing the test setup are presented in Fig. 16. The superstructure is intended as the linear SDOF system while the substructure is meant to be fixed. The level of material and structural complexity is low and the probability of existing damage is unlikely, so the reliability of this model as a linear SDOF system is considered to be high except possibly for the question of base fixity. Two accelerometers (Silicon Designs 2210-002) are mounted to the top and bottom masses, while the difference in accelerometer readings is what is to be analyzed in order to overcome the base fixity issue.

The fundamental frequency of the SDOF system is estimated as $f_n = 2.74$ Hz, or $\omega_n = 2\pi f_n = 17.2$ rad/s, based on the following material properties and geometric details:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

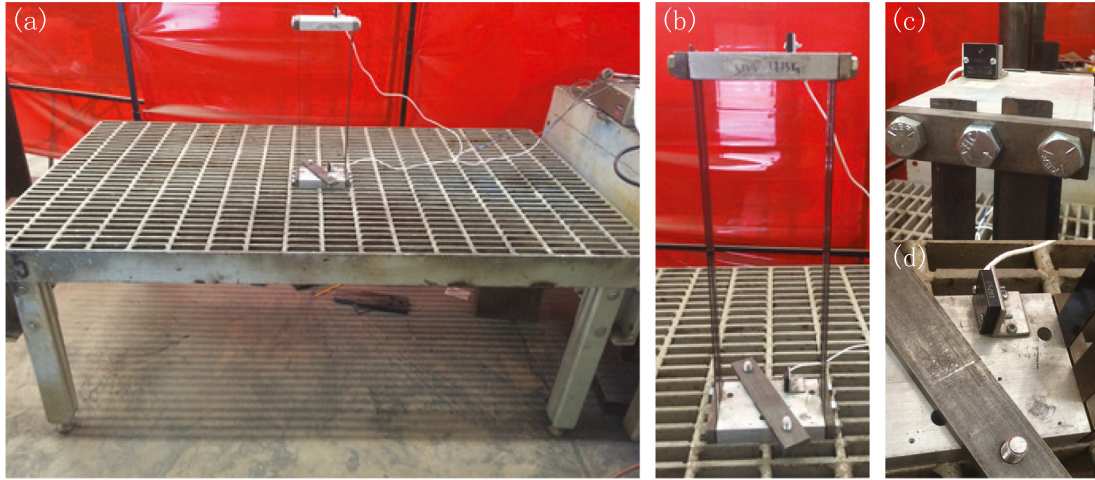


Fig. 16. Test setup, where (a) is an overview; (b) is the superstructure; (c) is the accelerometer mounted on the top mass, and (d) is the accelerometer mounted on the bottom mass.

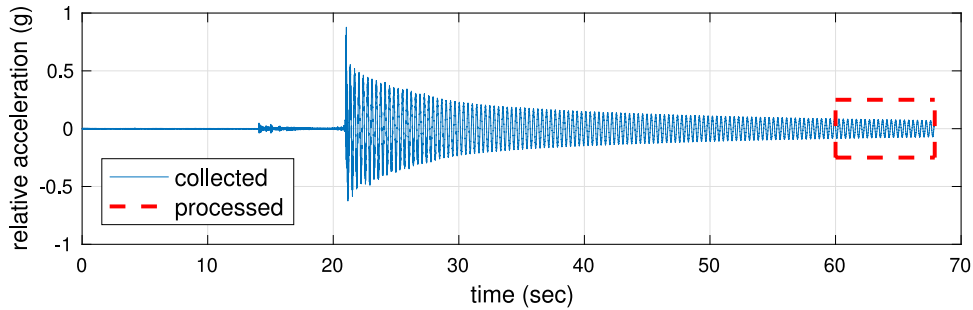


Fig. 17. The relative accelerometer data and the portion (in red box) to be processed.

where $mg = 4.1276$ lbf, and k is estimated as follows, where k stands for the stiffness of the entire model, contributed by four identical columns, each of which is a thin metal plate, and $\bar{k}$ stands for one such column:

$$k = 4\bar{k}, \quad \bar{k} = \frac{12EI}{h^3}$$

with Young's modulus $E = 22.324 \times 10^6$ psi (from independent laboratory tests), and

$$I = \frac{1}{12}bd^3, \quad \text{with } b = 1.0 \text{ in and } d = 0.0625 \text{ in}$$

Finally, the clearance is 18 in, while the width of each plate is 1 in. Thus, $h = 19$ in. This leads to the estimated $f_n = 2.74$ Hz.

In the laboratory test, the top mass of the superstructure was pulled horizontally with a wire, creating a non-zero displacement in the SDOF direction, and the system was allowed to come to rest in that position as measured by the two accelerometers. The wire was then slowly cut until the top mass suddenly released. Fig. 17 gives the entire time history of the collected relative acceleration of the top and bottom masses. The cutting of the wire began at (roughly) 14 s and continued until 22 s, at which time the relative acceleration jumped to nearly 1g. As indicated in Fig. 17 by the red box, only a small portion of the relative accelerometer data (from 60 to 67.9 s) was utilized for backbone extraction and statistical analysis, a brief summary of which is presented next.

Although a sampling rate of $f_s = 10$ kHz was used to capture the original accelerometer data, the portion within the red box (from 60 to 67.9 s) was downsampled by 100, resulting in an effective sampling rate of $f_s = 100$ Hz in all cases presented herein. A power spectral density calculation of this portion of the relative accelerometer data shows a dominant peak around 2.88 Hz, a value that agrees quite nicely with $f_n = 2.74$ Hz, the previously estimated fundamental frequency of the SDOF model.

The downsampled ($f_s = 100$ Hz) relative accelerometer data shown in Fig. 17 (red box) were searched, in order to identify sample points corresponding to local extrema. The first and last extrema (minima) were selected to act as zero-slope points at the each end, yielding a 766-point acceleration signal (7.65 s long) with 22 cycles. Backbones were

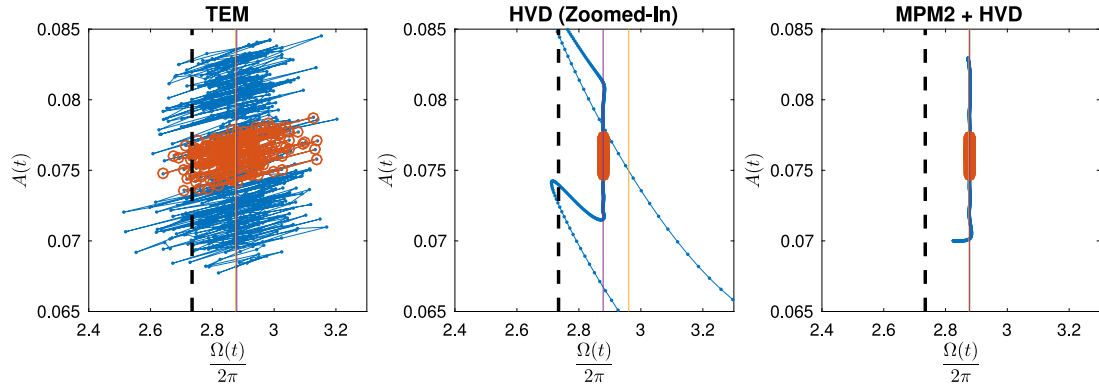


Fig. 18. Three approximate analytic-signal backbones: (a) MPM+hilbert.m, (b) HVD, (c) MPM2+HVD; extracted using 766-point relative acceleration signal in Fig. 17 (red box). The portion from 63 to 64.6 s is highlighted in each panel. The natural frequency identified by the FFT is shown as a black dashed line in each panel. HVD with $n = 1$, $f_p = 0.02$ (user-defined) in “synchdem.m” and “lpf.m”, $f_p = 0.01$ (built-in) in “hilbfir.m”.

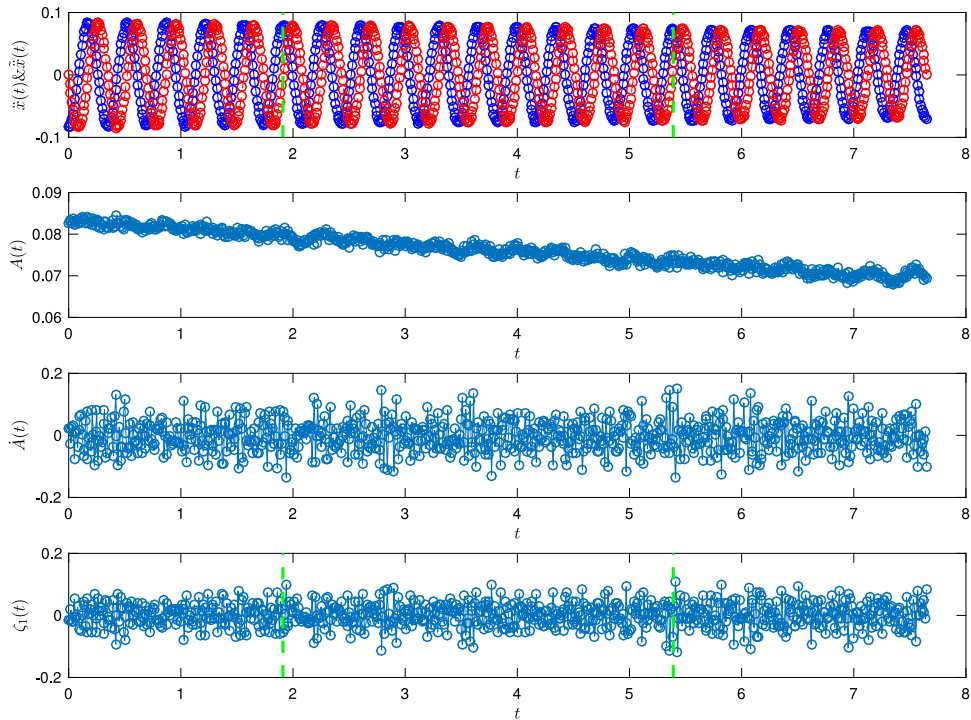


Fig. 19. Example of estimating ζ by using MPM+hilbert.m and averaging $\zeta_1(t)$ over the middle 10 cycles (out of 22 cycles) of the relative acceleration signal in Fig. 17 (red box).

extracted in three ways, using MPM+hilbert.m, HVD and MPM2+HVD, the results of which are presented in Fig. 18. For clarity, an interior portion (from 63 to 64.6 s) is highlighted in each panel, and the average frequency of each backbone together with the natural frequency identified by FFT are shown in Fig. 18. Corresponding to the same three methods of backbone extraction, Figs. 19–21 show three examples of estimating the damping ratio ζ by averaging $\zeta_1(t)$. Finally, Fig. 22 shows histogram distributions of estimated ζ values by using ensembles of acceleration signals in conjunction with each method (MPM+hilbert.m, HVD, and MPM2+HVD). All in all, this indicates that the SDOF model has a dominant mode at $f_n = 2.9$ Hz with damping ratio $\zeta = 1.4 \times 10^{-3}$.

8. Summary and conclusions

The straight-line backbone in Fig. 1 comes from a linear steady-state forced-vibration problem, not the transient free-vibration problem Eq. (3). Moreover, theoretical challenges arise when considering the Hilbert transform (HT) of

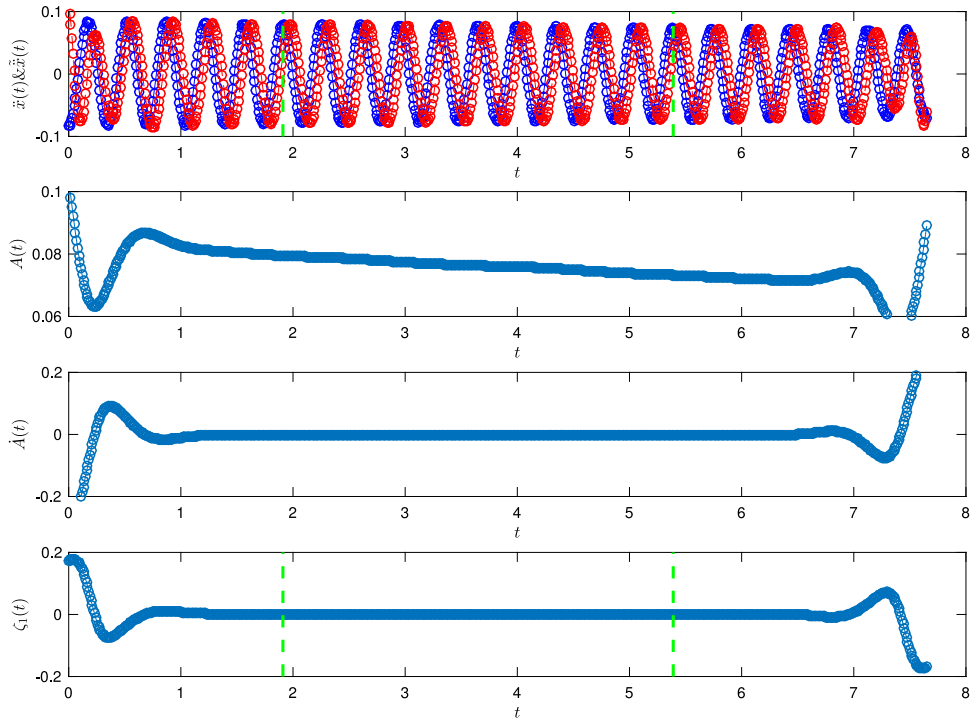


Fig. 20. Example of estimating ζ by using HVD and averaging $\zeta_1(t)$ over the middle 10 cycles (out of 22 cycles) of the relative acceleration signal in Fig. 17 (red box); HVD with $n = 1$, $f_p = 0.02$ (user-defined) in “synchdem.m” and “lpf.m”, $f_p = 0.01$ (built-in) in “hilbfir.m”.

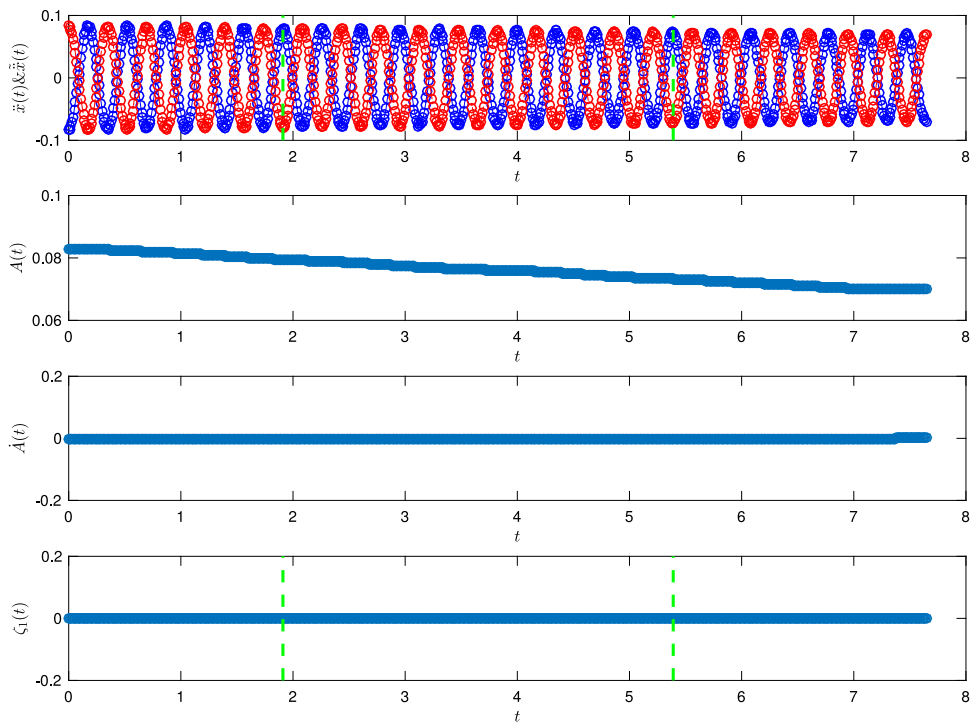


Fig. 21. Example of estimating ζ by using MPM2+HVD and averaging $\zeta_1(t)$ over the middle 10 cycles (out of 22 cycles) of the relative acceleration signal in Fig. 17 (red box); HVD with $n = 1$, $f_p = 0.02$ (user-defined) in “synchdem.m” and “lpf.m”, $f_p = 0.01$ (built-in) in “hilbfir.m”.

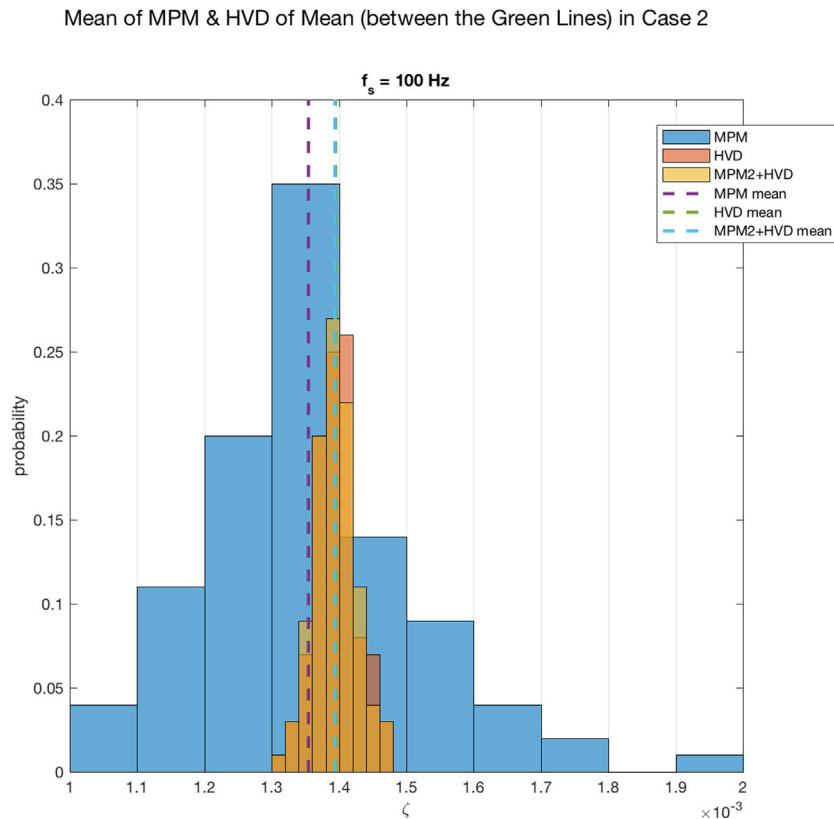


Fig. 22. Histogram distributions of estimated ζ values using acceleration-signal ensembles and three methods of backbone extraction, MPM+hilbert.m, HVD, and MPM2+HVD.

damped solutions of Eq. (3) because such transforms do not exist. Hence it is unclear why analytic-signal backbones of Eq. (3) should match the linear-theory backbone in Fig. 1. However, the Mirror Periodic Method (MPM) provides a practical starting point for dealing with end effects when approximating analytic-signal backbones for lightly damped structures. In particular, as the number of damped cycles increases in simulations, MPM backbones to oscillate around the physically desired straight-line AM backbones. Although unfiltered MPM backbones are oscillatory rather than straight, filtering can (in principle) reduce the oscillations to an acceptable level, especially if damping is light, even when analyzing accelerometer data. In the spirit of reproducible computational research [27], MATLAB code corresponding to the simulation results in Figs. 3–15 will be provided upon request to the corresponding author.

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